



MYCELIAL GRAPH V1

Research Edition

Architecture, frozen experimental protocol, demonstrable POC, and
evidence-gated evolution plan

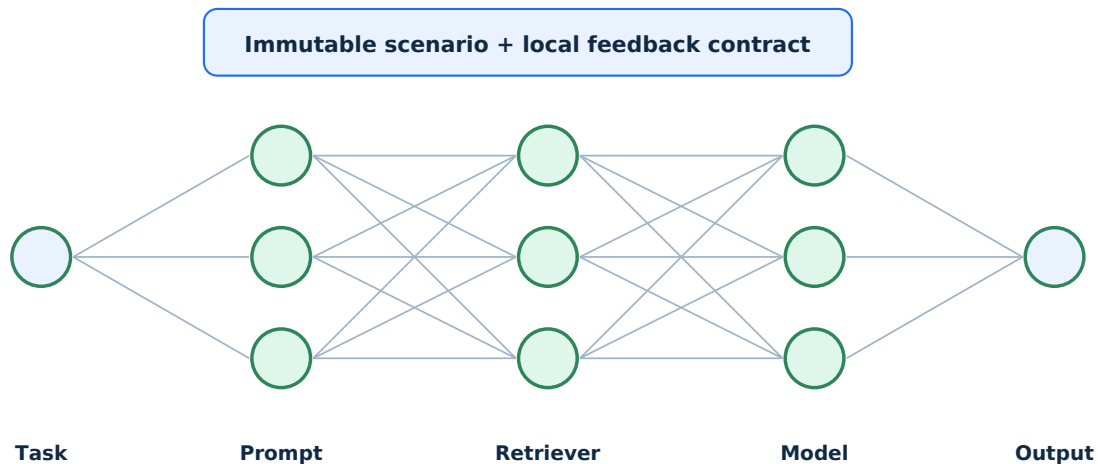
Crizan Belem Ribeiro
Independent Researcher | Sao Paulo, Brazil

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Executive overview

Mycelial Graph is a research framework for adaptive routing across heterogeneous AI execution components. Its motivating idea is simple: when a local provider, model, retriever, tool, or cloud component degrades, the system should update the affected region while preserving useful state elsewhere.



The biological metaphor provides design intuition, not proof. V1 converts it into a falsifiable experiment: hierarchical node-edge state should recover with fewer observations when a shock is shared through a node, but it must not create unacceptable negative transfer when a disruption belongs to one edge.

What V1 is	What V1 is not
A local, reproducible simulator with frozen paired scenarios, four methods, paired analysis, automatic reporting, and a locked confirmatory workflow.	A production multicloud router, a proof of convergence, a provider benchmark, or evidence that hierarchy is already superior.

1. The original idea, preserved

Production AI systems are graphs, even when their code looks like a linear pipeline. A task can select among prompt strategies, retrieval services, models, tools, guardrails, parsers, regions, and fallbacks. Costs, latency, quality, quotas, and availability change independently.

The V0 contribution was an adaptive edge conductance: useful local flow reinforces a connection; inactivity and changing evidence allow it to decay. V1 keeps this mechanism intact as the edge-only control. The project does not replace its core idea with generic reinforcement learning, a graph neural network, or a large orchestrator.

V1 adds one candidate mechanism: partial pooling through node state. This is a scientific extension, not a declared product upgrade.

Core invariant

LOCAL ADAPTATION + STRUCTURAL REUSE + HARD EVIDENCE BOUNDARIES

Product objective

Minimize cost per successful task while preserving explicit quality, latency, reliability, safety, and policy requirements. Token price alone is never the optimization target.

Scientific objective

Measure whether the recovery burden changes as the true shared fraction of a localized shock changes, and identify when sharing becomes harmful.

Engineering objective

Create a small local package that can later accept frozen real-provider traces and bounded live adapters without rewriting its experimental core.

2. The research architecture

Each module has one responsibility. The environment produces an immutable world. Methods choose and update using only permitted local feedback. The runner pairs results. Analysis consumes raw records but cannot change them.

Layer	Responsibility	Invariant
Configuration	Frozen graph, horizon, shock, methods, metrics, and seeds	YAML is the single scientific source of truth
Environment	DAG, latent means, indexed potential outcomes, certified optima	Read-only after generation
Methods	Choose a path and update local state	Same observation contract; isolated RNG
Runner	Execute paired scenarios, traces, and checkpoints	Atomic, idempotent, serial/parallel equivalent
Analysis	RRT, RMST estimate, regret, bootstrap, gates	No tuning or seed filtering
Reporting	Tables, figures, interpretation boundary	Run kind always visible

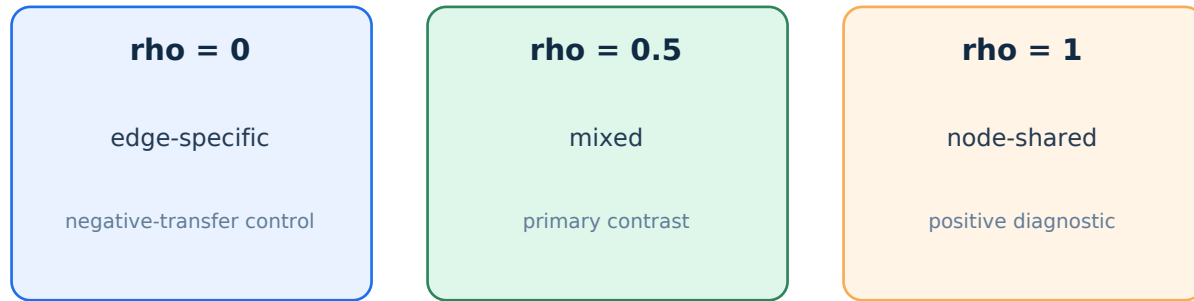
Immutable scenario contract

Potential outcomes are indexed by step and edge. If two methods choose the same edge at the same time, they receive the same local noise realization. Different actions query different outcomes from the same frozen world. This is stronger than reusing one mutable RNG sequence.

Randomness

Stable namespaces derive independent streams for scenario parameters, local outcome noise, and each method. Completion order under ProcessPoolExecutor cannot change the canonical scientific payload.

3. Controlled structural sharing



Total L2 shock magnitude remains constant

The experiment constructs orthogonal node and interaction shock patterns. The parameter ρ reallocates squared shock magnitude between them while total L2 magnitude remains fixed.

```
d(rho) = -m [ sqrt(rho) n + sqrt(1-rho) i ]
```

For each seed, all ρ values share the same graph, pre-shock edge means, and potential-noise table. A candidate family is admitted only when every configured ρ has a changed, unique post-shock optimum with the minimum utility margin.

Why the endpoints matter

- $\rho=0$ is the negative-transfer control: a hierarchy that broadcasts an edge-specific observation can hurt.
- $\rho=1$ is the positive diagnostic: fully shared node information is the most favorable pooling regime.
- $\rho=0.5$ is the primary mixed regime: shared and specific components contribute equally in squared magnitude.

What methods observe

Methods receive local rewards only for traversed edges. Expected utility, shock location, future outcomes, and oracle paths remain evaluation-only information.

4. Methods and fair comparison

Method	State	Selection	Scientific role
MG edge-only	Independent bounded edge conductance	Softmax + basal exploration	Preserves the V0 mechanism
Node-only	Source and target node effects	Same MG policy family	Pooling ablation
MG hierarchical	Source + target + edge interaction	Same MG policy family	Proposed partial pooling
Structured SW-UCB	The same node-edge feature family	Sliding-window UCB	Strong representation-aware baseline

Hierarchical score

$$\text{score}(u,v,t) = \text{base} + a[u,t] + b[v,t] + c[u,v,t]$$

Online updates use sum-to-zero projection and shrinkage. These constraints make the numerical parameterization reproducible, but they do not manufacture evidence for rarely observed components.

Why SW-UCB is mandatory

Without a baseline that uses the same shared features, a hierarchical victory could be attributed only to representation. SW-UCB helps distinguish representation value from the specific Mycelial policy dynamics.

Oracle boundary

The oracle knows expected utilities and defines the optimum. It does not compete, receive confidence intervals, or represent a deployable method. Its realized reward need not beat every stochastic draw.

5. Recovery and decision gates

Recovery is not one lucky optimal choice. It requires a trailing mean of expected selected-path utility to reach at least 90% of the post-shock oracle and remain there through a confirmation window.

$$\text{RRT}_i = \min(\text{T_recovery}_i, \tau)$$

The mean individual RRT estimates restricted mean time without recovery when censoring is administrative at τ . Lower is better.

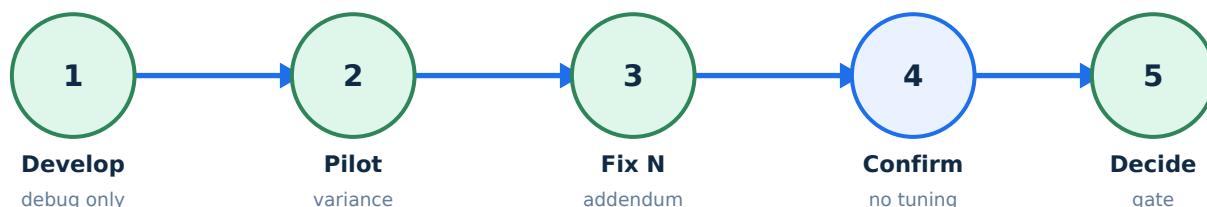
Primary relative effect

$$\text{delta} = [E(\text{RRT_hierarchical}) - E(\text{RRT_edge-only})] / E(\text{RRT_edge-only})$$

Gate	Frozen rule	Meaning
Superiority	95% one-sided upper bound < 0 at $\rho=0.5$	Evidence of lower recovery burden
Engineering value	Point estimate $\leq -20\%$	Benefit may justify added complexity
Negative-transfer safety	Upper bound $< +10\%$ at $\rho=0$	No unacceptable edge-specific harm
Promotion	All scientific gates + operational cost budget	Hierarchy may advance to product POC

A non-significant result is inconclusive unless the experiment had enough power to exclude the relevant benefit. Pilot and development gates are diagnostic only.

6. Reproducible execution lifecycle



The confirmatory YAML deliberately references a seed file that does not yet exist. After the pilot, the sample-size calculation is recorded in an addendum and the first N seeds are copied mechanically from a precommitted 500-seed pool.

Required artifacts

Artifact	Purpose
EXPERIMENT_PROTOCOL_V1.md	Frozen hypotheses, observation contract, metrics, gates, and prohibitions
ANALYSIS_PLAN.md	Estimand, paired bootstrap, censoring, multiplicity, and reporting
SAMPLE_SIZE_ADDENDUM.md	Pilot variance, power method, N, dates, and selected seed range
experiment.schema.json	Machine-readable paired result contract
manifest.json	Config, seed, code, dependency, and artifact hashes
REPORT.md	Automatic result and interpretation boundary

Checkpoint behavior

Completed checkpoints are reused only when seed, rho, config hash, and code revision agree. A conflicting output is never silently overwritten. Traces use atomic temporary-file replacement.

7. Sustainable project design

Sustainability means keeping the project scientifically honest, technically maintainable, financially bounded, and computationally proportionate.

Dimension	V1 commitment
Scientific	Immutable raw data, disjoint phases, amendments, and explicit negative-result paths
Technical	Typed small modules, local files, no database/cloud/orchestrator dependency
Financial	Synthetic evidence before paid calls; one-cloud canary before multicloud
Computational	Power-justified N, parallel local trials, idempotent checkpoints, CPU reporting
Governance	Cost optimization never overrides hard safety, privacy, or compliance constraints

Why there is no orchestrator yet

Independent trials map cleanly to ProcessPoolExecutor. An external workflow engine becomes justified only when local duration, multi-node compute, dependency complexity, or infrastructure recovery becomes a measured bottleneck.

Why there is no SQLite yet

Twenty to a few hundred paired scenarios remain transparent as JSON plus compressed traces. A database adds value only when cross-version querying becomes a repeated requirement.

Why there are no live prices

Provider values are temporally unstable. Future real-world work must capture dated immutable traces before replay; live queries during a confirmatory run would break reproducibility.

8. Evidence-gated evolution

Stage	Question	Deliverable	Advance only if
V1 research	Does partial pooling help under shared shocks?	Confirmatory report	Scientific gates are evaluated
Single-cloud POC	Does the effect survive realistic traces and bounded calls?	Cost/quality/latency case	V1 supports a useful regime
Composite pipeline	Does graph routing help beyond model choice?	Retriever/tool/guardrail POC	Single-cloud value is measured
Multicloud fabric	Does distribution solve a real resilience or sovereignty need?	Governed execution platform	One cloud is demonstrably insufficient

First cloud selection

- Prefer credits, low setup cost, version pinning, observable telemetry, and strict budgets.
- Require at least two meaningful routing alternatives behind one narrow task contract.
- Replay frozen traces locally before any live canary.
- Measure cost per successful task, quality, p95 latency, failure rate, and recovery.

Deferred research

Dynamic graph discovery, distributed consensus, causal attribution, counterfactual explanations, 10,000-node scaling, and formal regret bounds remain valid future questions. None is required to answer MG-EXP-V1.

9. Demonstrable POC and current status

The repository includes an end-to-end development command that validates configuration, generates paired scenarios, executes all four methods, calculates paired metrics, creates plots, writes a manifest, and produces a final Markdown report.

Windows PowerShell

```
python -m venv .venv
Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned
& .\.venv\Scripts\Activate.ps1
python -m pip install -e .
mycelial-graph demo
```

Output contract

```
outputs/demo/
raw/ - paired immutable scenario files
traces/ - compressed step decisions and selected scores
processed/analysis.json - aggregate estimates and gates
figures/ - recovery and regret plots
manifest.json - environment and hashes
REPORT.md - final human-readable result
```

Evidence status

DEVELOPMENT-ONLY

The included output verifies execution and can falsify naive expectations, but it is not a confirmatory scientific result. The confirmatory configuration remains locked until the independent pilot determines N.

10. Definition of done

V1 is complete when the evidence cycle closes, not when the repository accumulates features.

- All deterministic, pairing, shock-magnitude, censoring, and checkpoint tests pass.
- The independent pilot is executed once and N is recorded transparently.
- Confirmatory seeds are selected mechanically from the frozen pool.
- No tuning or adaptive intervention occurs during confirmatory execution.
- The automatic report includes passed and failed gates, uncertainty, limitations, and failures.
- The decision is documented even when the hypothesis is unsupported or inconclusive.

The project remains faithful to its origin by treating local adaptation as a mechanism to test, not a metaphor to protect. Its capacity to grow comes from stable contracts and evidence gates, not from adding every future feature to the first implementation.

NEXT ACTION

Review the frozen artifacts, run the development POC locally, then execute the independent pilot. Do not unlock the confirmatory configuration before the sample-size addendum is committed.